

Mahavir Chaudhari

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Experience

Data Engineer II, Providence Global Center - Hyderabad Jul 2025 - Present

- **Designed and implemented a financial reconciliation data layer in Snowflake** using dimensional modeling, SCD Type 2 dimensions, ELT pipelines, and batch processing, consolidating **1B+ records daily** into analytics-ready datasets, reducing manual reconciliation effort by **80%** while enabling audit-ready financial reporting.
- **Architected the AI Data Lifecycle Companion (AIDLC)**, an enterprise platform for agentic AI that standardizes **context engineering**, MCP integrations, reusable skills, agent orchestration, and HIPAA-compliant guardrails, enabling Data Engineering teams to rapidly build scalable, production-grade AI applications.
- **Modernized** enterprise data infrastructure by migrating **400+ PySpark ETL workloads and 11 production ingestion pipelines** from **Databricks to Snowflake**, leveraging AI agents, reusable skills, and automation tools to accelerate migration while ensuring functional parity and **zero production downtime**.

Data Engineer I, Providence Global Center - Hyderabad Jul 2023 - Jun 2025

- **Re-architected** the Mandatory Days Off (MDO) scheduling engine by redesigning data models, query paths, and batch workflows across enterprise warehouses and curated data marts, reducing execution time from **2 hours to 10 minutes (92%)** while supporting scheduling for **6,000+ clinicians**.
- **Engineered** a metadata-driven validation framework using **Python, Streamlit, and Dynamic SQL generation**, implementing reusable validation rules, schema verification, and automated data reconciliation across production ETL pipelines, reducing QA execution time by **86% (7 hours → 1 hour)** while improving data reliability.

Software Development Intern, Fortranix Technologies - Bangalore Jan 2023 - Jun 2023

- **Developed** automated **Rust** unit tests for an enterprise software security platform, increasing code coverage from **30% to 80%** and strengthening production reliability through comprehensive validation.

Software Developer (Part-time), Streamn - Remote May 2021 - Aug 2022

- **Built** GPU-accelerated computer vision models using Python, TensorFlow/PyTorch, Jupyter, and Google Colab to automate logo detection across video streams, improving the scalability of media content analysis.

Projects

Enterprise AI Context Engineering

- **Architected** a containerized **context engineering platform** for enterprise AI agents using Python, Docker, MCP, and LLMs, implementing a layered context architecture with **semantic layers, ontologies, knowledge graphs**, and reusable tools to integrate enterprise systems and standardize AI-assisted data engineering workflows.
- **Engineered** a modular multi-agent runtime with **15+ reusable Data Engineering skills**, dynamic context composition, semantic retrieval, tool orchestration to Oracle, Snowflake, and ADF, and enterprise **HIPAA-compliant guardrails**, enabling complex KTLO operations through natural language.

Smart Research Scout

- **Designed and developed a full-stack AI research platform** using FastAPI, React, PostgreSQL, Redis, and vector search, aggregating technical knowledge from research papers, developer communities, and social platforms into a unified **semantic search and analytics dashboard**.

Technical Skills

- **Data Engineering:** Python, SQL, Snowflake, Azure Data Factory (ADF), Distributed systems, PySpark, dbt, ETL/ELT, Data Modeling, Dimensional Modeling, SCD, Data Warehousing, Batch Processing, Control-M
- **AI Systems:** Agentic AI, Context Engineering, Semantic Layers, Ontologies, Knowledge Graphs, Retrieval-Augmented Generation (RAG), MCP, LangGraph, AI Agents, Tool Calling, Semantic Search, Vector Search, APIs, Orchestration
- **Cloud:** Azure, AWS, GCP, Docker, Git, GitHub Actions
- **Languages & Frameworks:** FastAPI, Flask, React, PostgreSQL, Redis, MySQL, SQLite, Java, Go, Rust, Bash, JavaScript, HTML/CSS

Education

BITS Pilani, BE in Computer Science

2019-2023